

Same plot, two different errors

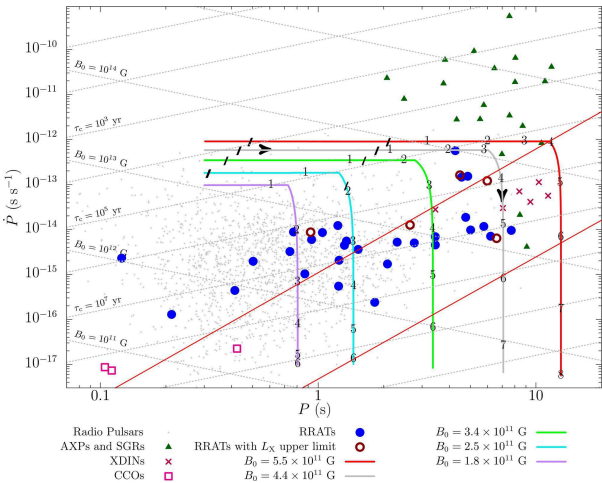
Question: Which neutron star group predominantly has a characteristic age ( $\tau_c$ ) less than  $10^5$  years according to the plot?

A. AXPs and SGRs B. RRATs C. XDINs D. Radio Pulsars

Human evidence check

Age contours are gray dotted diagonals.  
At a fixed period, positions above the  $10^5$ -year contour indicate younger stars; positions below indicate older stars.  
Most AXPs and SGRs (green triangles) lie above this contour.

Reference answer: A. AXPs and SGRs



Original plot

Misidentified age contour

Thinking before the tool (full)

To determine which neutron star group has a characteristic age ( $\tau_c$ ) less than  $10^5$  years, we need to focus on the region of the plot where  $\tau_c = 10^5$  years is indicated. This region is located in the lower part of the plot, below the  $\tau_c = 10^5$  yr line. Cropping and zooming into this specific area will help us identify the relevant data points and their corresponding labels.

The task involves analyzing the distribution of data points in the lower section of the plot. Given the scale and resolution of the image, zooming in on the area below the  $\tau_c = 10^5$  yr line will provide a clearer view of the data points and their labels, making it easier to identify the predominant group with  $\tau_c < 10^5$  years.

Here is the Python code to crop and zoom into the relevant area:

Reversed age comparison

Thinking before the tool (full)

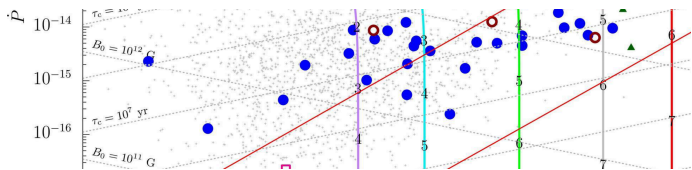
To identify which group of neutron stars primarily has a characteristic age ( $\tau_c$ ) less than  $10^5$  years, we need to concentrate on the area of the plot where  $\tau_c = 10^5$  years is indicated. This region is positioned in the middle of the plot, with  $\tau_c$  values diminishing toward the left side.

Focusing on this specific area by cropping and zooming into it will facilitate the clear identification of data points associated with each neutron star category (AXPs and SGRs, RRATs, XDINs, and Radio Pulsars) and their respective  $\tau_c$  values. This approach aids in accurately determining which category predominantly falls below  $10^5$  years.

Here is the Python code to crop and zoom into the relevant area:

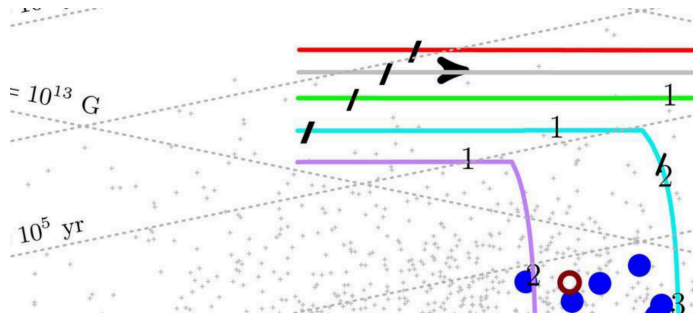
Actual tool crop

Crop (0, 700) to (1932, 1150); enlarge by 1.5×



Actual tool crop

Crop (300, 400) to (1200, 800); enlarge by 1.5×



Tool code is summarized above; all natural-language thinking is retained.

Thinking after the tool (full)

The green lines on the plot correspond to  $\tau_c = 10^5$  years, and most of the RRATs (represented by blue circles) are found below these lines, indicating that they predominantly have a characteristic age less than  $10^5$  years.

Thinking after the tool (full)

The plot features lines depicting characteristic ages ( $\tau_c$ ) for different magnetic field strengths. AXPs and SGRs are primarily located above the  $\tau_c = 10^5$  years line, indicating that their characteristic ages exceed  $10^5$  years. Conversely, RRATs, marked by blue circles, are predominantly situated below the  $\tau_c = 10^5$  years line, suggesting that their characteristic ages are generally less than  $10^5$  years.

× Final answer: B. RRATs

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